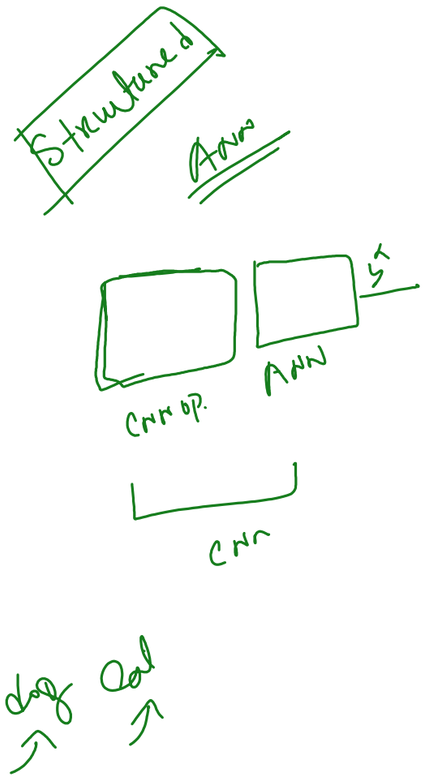
What is a CNN?

CNN (Convolutional Neural Network) is a type of neural network designed to **understand images**.



One-line definition

"A CNN is a neural network that automatically learns important features from images using filters."

Intuition

- Humans see → shapes, edges, objects
- CNN sees → **numbers (pixels)** and learns patterns

How CNN thinks

A CNN learns in stages:

1. **Edges** (lines, curves)
2. **Textures** (patterns)
3. **Parts** (eyes, wheels)
4. **Objects** (face, car, dog)

Core idea

CNN uses **filters (kernels)**

- A small matrix slides over the image
- Extracts important features

Main components

1. Convolution Layer

- Detects features (edges, patterns)

2. Pooling Layer

- Reduces size (keeps important info)

3. Fully Connected Layer

- Makes final decision (classification)

Example

Input:

- Image of a cat 😸

  Output:

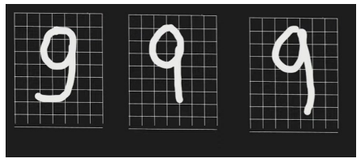
  ```
  Cat (95%)
  Dog (5%)
  ```

Simple analogy

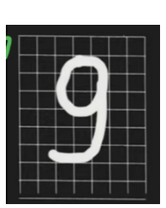
"CNN is like a smart eye that first sees edges, then shapes, then understands the object."

Final takeaway

- CNN = works on images
- Input = pixel values
- Output = prediction

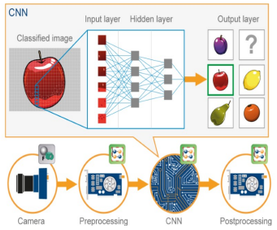


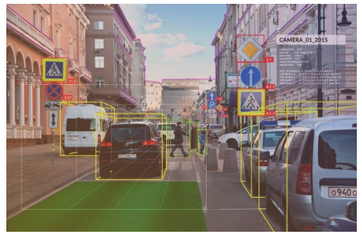
intuition



CNN Intuition :

CNN





🖼️ 1. Image Classification

👉 Identify what is in an image

Examples:

- Cat vs Dog
- Disease detection from leaf images (your project)

🔍 2. Object Detection

👉 Find **what + where** in an image

Examples:

- Detect cars, people in traffic
- Security surveillance systems

🎯 3. Image Segmentation

👉 Classify **each pixel**

Examples:

- Medical imaging (tumor detection)
- Self-driving cars (road, lane detection)

😊 4. Face Recognition

👉 Identify a person

Examples:

- Phone face unlock
- Attendance systems

✍️ 5. Handwritten Digit Recognition

👉 Read handwritten text

Examples:

- Postal code reading
- Bank cheque processing

Our Application :

🚗 6. Self-Driving Cars

👉 Understand surroundings

Examples:

- Detect lanes
- Recognize traffic signs

🏥 7. Medical Image Analysis

👉 Assist doctors

Examples:

- X-ray analysis
- MRI/CT scan diagnosis

🎥 8. Video Analysis

👉 Apply CNN frame-by-frame

Examples:

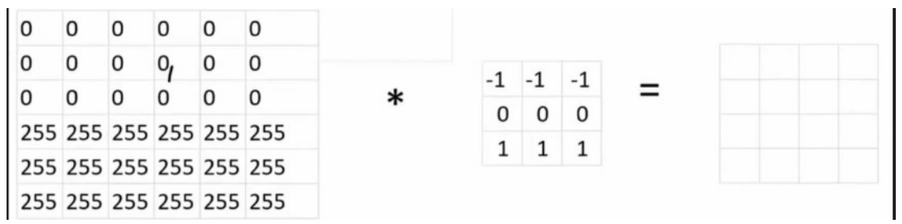
- Action recognition
- CCTV monitoring

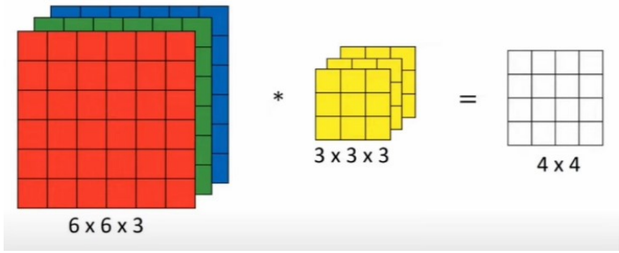
🎨 9. Image Generation & Enhancement

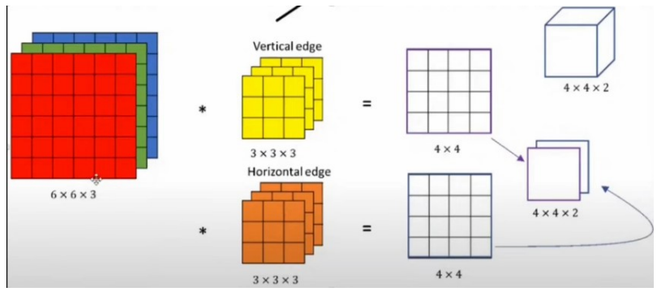
👉 Create or improve images

Examples:

- Super-resolution
- Style transfer




*
=
3 x 3 x 3
4 x 4
6 x 6 x 3


Vertical edge
6 × 6 × 3
3 × 3 × 3
4 × 4
4 × 4 × 2
Horizontal edge
3 × 3 × 3
4 × 4
4 × 4 × 2